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Annexe 2
Appendix 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2.27
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Methane ($\lambda \approx 3.39\ \mu\mathrm{m}$) (2003)

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Appendix 2

Absorbing molecule CH_4,$F_2^{(2)}$ component, P(7) ν_3 transition

1. CIPM recommended values

1.1.

The values $f = 88\,376\,181\,600.18\ \mathrm{kHz}$
$\lambda = 3\,392\,231\,397.327\ \mathrm{fm}$

with a relative standard uncertainty of 3×10^{-12} apply to the radiation of a He-Ne laser stabilized to the central component, (7-6) transition, of the resolved hyperfine-structure triplet. The values correspond to the mean frequency of the two recoil-split components for molecules which are effectively stationary, i.e. the values are corrected for second-order Doppler shift.

1.2.

The values $f = 88\,376\,181\,600.5\ \mathrm{kHz}$
$\lambda = 3\,392\,231\,397.31\ \mathrm{fm}$

with a relative standard uncertainty of 2.3×10^{-11} apply to the radiation of a He-Ne laser stabilized to the centre of the unresolved hyperfine-structure of a methane cell, within or external to the laser, held at room temperature and subject to the following conditions:

- methane pressure ≤ 3 Pa;
- mean one-way intracavity surface power density (i.e., the output power density divided by the transmittance of the output mirror) $\leq 10^4\ \mathrm{W\ m^{-2}}$;
- radius of wavefront curvature ≥ 1 m;
- inequality of power between counter-propagating waves $\leq 5\%$;
- servo-referenced to a detector placed at the output facing the laser tube.

2. Source data

2.1.

Resolved hyperfine structure

Adopted value $f = 88\,376\,181\,600.18(27)$ kHz
$u_c/y = 3 \times 10^{-12}$

for which:
$\lambda = 3\,392\,231\,397.327(10)$ fm
$u_c/y = 3 \times 10^{-12}$

calculated from

x/kHz	Laser	Frequency chain	Year	source data
600.29	LPI	PTB	1991	[1]
599.9	LPI	VNIIFTRI	1985-1986	[2]
600.11	LPI	VNIIFTRI	1989-1992	[2]
600.18	PTB	VNIIFTRI	1989	[2]
600.16	PTB	PTB	1992	[3]
600.44	ILP	ILP	1988-1991	[4]
Unweighted mean		$f = 88\,376\,181\,600.18$ kHz		

where $f = (88\,376\,181\,000 + x)$ kHz.

Other available values having uncertainties larger than 200 Hz have not been used. The relative standard uncertainty of one measurement was estimated to be 2.9×10^{-12} using the maximum deviation from the mean and rounded to 3×10^{-12}.

2.2.

Unresolved hyperfine structure

Adopted value $f = 88\,376\,181\,600.5(2.0)$ kHz
$u_c/y = 2.3 \times 10^{-11}$

for which:
$\lambda = 3\,392\,231\,397.31(8)$ fm
$u_c/y = 2.3 \times 10^{-11}$

calculated from

x/kHz	Frequency source	Frequency chain	Year	source data
600.9	Stationary device	ILP	1983	[4], [5], [6], [7]
601.48	Portable laser 2	NRC	1985	[8], [9]

x/kHz	Frequency source	Frequency chain	Year	source data
599.33	Portable laser 3	NRC	1986-1991	[8], [9]
596.82	Portable laser 1	AIST	1988-1990	[9]
601.52	CH_4 beam	PTB	1987-1989	[9], [10], [11]
601.77	Portable laser M101	VNIIFTRI	1985-1992	[2], [9]
600.12	Portable laser P1	VNIIFTRI	1985-1988	[2], [9]
598.5	Portable laser PL	VNIIFTRI	1986	[2]
600.96	Portable laser B.3	BIPM	1985-1992	[9]
601.33	Portable laser VB	BIPM	1988-1991	[9]
600.3	Portable laser VNIBI	BIPM	1991	[9], [10], [11], [12]
Unweighted mean:		$f = 88\ 376\ 181\ 600.46$ kHz		
where $f_{CH4} = (88\ 376\ 181\ 000 + x)$ kHz.				

The standard deviation of one determination is 1.7 kHz. This is equivalent to a relative uncertainty of 1.9×10^{-11}, increased by the CCL to 2.3×10^{-11} to give an uncertainty of 2 kHz.

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